

View Review

Paper ID

35

Paper Title

Cost-Benefit Analysis of Serverless Computing for Large-Scale AI Workloads in Content Generation

REVIEW QUESTIONS

1. Reviewer's confidence.

High

2. Does the paper describe the original work

Average

3. Does the work add value to the research of relevant areas?

High

4. Please write detailed Comments for - Suitability of Title and quality of the abstract. - Adequacy of literature review and proposed methods -Quality of result analysis and conclusion .

The title is clear about the technologies and use case discussed serverless computing for AI content generation and appropriately captures the breadth of the research. The motivation, methods, main conclusions, and practical significance are succinctly explained in the abstract. It is instructive and well-written.

- Adequacy of literature review and proposed methods

The extensive literature analysis includes a number of current investigations conducted between 2020 and 2024. It highlights pertinent research gaps and reviews previous findings on cost, scalability, and latency (e.g., cold-start difficulties, hybrid architectures, absence of AI-specific serverless optimization). More consideration of more current hybrid cloud-native solutions, such as serverless containers (like AWS Fargate) or Kubernetes with autoscaling, which are becoming more and more relevant alternatives, would enhance the assessment.

The process is reliable and repeatable. It does comparison testing on AI tasks like text and image generation and makes use of realistic AWS and Azure settings. There is a comprehensive explanation of the task design, monitoring metrics, and TCO components.

-Quality of result analysis and conclusion

Cost breakdowns, scalability measurements, latency information, and resource usage comparisons are all included in the results; simple presentation. Diagrams and tables improve the clarity. With serverless approaches showing better for dynamic and changeable workloads but less cost-effective for continuous, long-running operations, the analysis effectively illustrates the trade-offs between cost and flexibility.

The findings are skillfully summarized in the conclusion, which also offers practical advice like the suggestion for hybrid designs. The empirical arguments would be strengthened.